

Network Incident Analysis: DNS Service Unavailability

Simulated Case Study

Incident Summary

Customers were unable to access yummyrecipesforme.com and received an ICMP Destination Unreachable response indicating that UDP port 53 was unreachable. Network traffic captured during the incident shows that DNS queries sent to the DNS server over UDP port 53 were unsuccessful. Instead of returning the requested IP address, the server returned an ICMP Destination Unreachable response stating that UDP port 53 was unreachable.

Because the DNS request could not be completed, the client was unable to resolve the domain name to an IP address and could not proceed with an HTTPS connection to the website.

Technical Analysis and Working Hypothesis

The issue was reproduced during the investigation while capturing network traffic with tcpdump. The client at 192.51.100.15 sent a DNS A-record query to the DNS server at 203.0.113.2 over UDP port 53. Instead of returning the website's IP address, the server responded with an ICMP Destination Unreachable message indicating that UDP port 53 was unreachable. The client retried the DNS request three times and received the same response each time.

Because DNS resolution failed, the client could not obtain the website's IP address and could not continue to the HTTPS connection. The traffic suggests that the DNS service was unavailable or not listening on UDP port 53, although the underlying cause still needs to be confirmed.

Next steps should include checking that the DNS service is running and listening on port 53, reviewing the server logs and configuration, and confirming that firewall rules are not blocking or rejecting DNS traffic.